

Western Blotting

Overview

To separate proteins based on their molecular weight using polyacrylamide gel electrophoresis.

A. Materials

Item	Amount (per gel)
SDS-PAGE sample gel	1 sheet
Power supply	1 unit
Wet transfer tank	1 unit
Shaker	1 unit
Chemiluminescence imager	1 unit
Micropipettes	As required
PVDF membrane (Millipore)	1 sheet
3M filter papers	2 sheets
Sponge pad	2
Cassette	1 unit
1X Transferring buffer	As required
His-Tag Antibody (AD1.1.10) (sc-53073, Santa Cruz Biotechnology)	As required
Peroxidase AffiniPure [®] Goat Anti-Mouse IgG (H+L) (Jackson ImmunoResearch)	As required
T-Pro LumiDura Chemiluminescent Substrate (ECL) Kit (T-Pro Biotech)	1 unit

B. Procedure

● Gel collection

1. Prepare protein samples in SDS sample/loading buffer.
2. Denature the proteins by heating at 95°C for 5–10 minutes.
3. Spin down briefly in a microcentrifuge to gather condensate..
4. Load 30-50 µg of total protein into each well.
5. Run the gel under appropriate electrophoresis conditions:
 - a. Stacking: Apply 80–120 V until samples pass into the resolving gel.
 - b. Separation: Increase to 100–200 V until the dye front reaches the gel bottom.

● Transferring

Assemble the transfer sandwich as following

6. Submerge the PVDF membrane in 100% methanol for 1 minute, then soak in 1X Transfer Buffer for 3 minutes.
7. Soak the polyacrylamide gel in 1X Transfer Buffer for 5 minutes.
8. Pre-wet the 3M filter papers in 1X Transfer Buffer

● Blocking

9. Block non-specific binding sites by incubating the membrane in 5% non-fat milk at room temperature for 1 hour.

● Antibody incubation

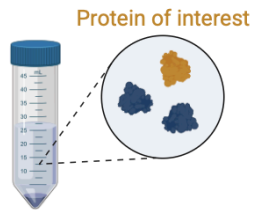
10. Incubate the membrane with primary antibody overnight at 4°C with continuous agitation.
11. Wash the membrane 3 times using 1X TBST.
12. Incubate with secondary antibody for 1 hour at room temperature with continuous agitation.
13. Thoroughly wash the membrane 5 times using 1X TBST.

● Detection

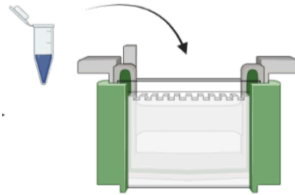
14. Combine ECL Reagents A and B in a 1:1 ratio.
15. Apply the mixed ECL reagent evenly over the membrane to detect target protein bands via chemiluminescence.

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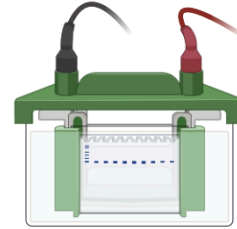
1 Protein extraction



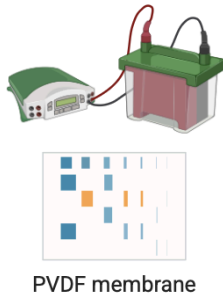
2 Load 30-50ug of total protein into each well



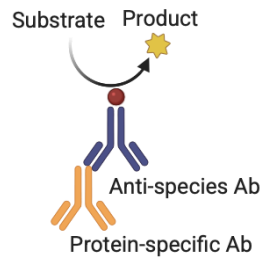
3 SDS page protein separation



4 Electrotransfer



5 Antibody probing



6 Chemi-imaging

